

Architecting a Business with MLOps and Data Science

From Models to Enterprise Learning Systems

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Most organizations invest heavily in Data Science, Machine Learning and MLOps, yet many struggle to turn analytical insights into measurable business value. This article argues that Data Science and MLOps should be treated as components of a broader enterprise architecture and that modern organizations must evolve from model-centric thinking toward *enterprise learning systems*: architectures capable of continuously sensing, learning, adapting and generating business value.

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1 Introduction

Most organizations invest heavily in Data Science, Machine Learning and MLOps across the data engineering, data governance and platform layers. In the particular case of models — classical classifiers, regressions, even forecasting models — models are trained, dashboards are built, experiments are executed, and cloud platforms process vast amounts of data every day. Yet despite these investments, many organizations still struggle to convert analytical insights into measurable business value.

One can ask what happens with the expected results, and the answer is usually the same: **the problem is not technological, the problem is architectural.**

To be concrete, machine learning models are often treated as isolated technical assets rather than as components of a larger enterprise system. Accuracy improves, yet decision quality remains unchanged. Experiments produce statistically significant results, yet business outcomes fail to improve. Monitoring platforms detect drift, yet strategic decisions continue to rely on outdated assumptions.

This article argues that Data Science and MLOps should be viewed as components of a broader enterprise architecture. Modern organizations must evolve from model-centric thinking toward **enterprise learning systems**: architectures capable of continuously sensing, learning, adapting and generating business value. Generative AI is, of course, also a standard to consider; for the purposes of this paper the same discussions apply.

2 The Problem with Traditional Data Science

Most machine learning initiatives optimize *models* instead of optimizing the *enterprise*. Teams frequently focus on:

- Prediction accuracy.
- AUC and F1 scores.
- R^2 , RMSE and similar regression metrics.
- Feature engineering.
- Hyperparameter tuning.

These metrics are important, but they are not business outcomes.

A highly accurate model with low adoption, poor operational integration or limited strategic impact generates little value. Business value emerges only when **predictions become decisions, decisions become actions and actions influence measurable outcomes**.

This observation leads to a simple but powerful principle:

The objective of Machine Learning is not prediction. The objective is **improved decision-making**.

3 The Enterprise as a (Dynamic) Learning System

A modern enterprise should not be understood as a static collection of processes, applications, dashboards and machine learning models. It is better understood as a **dynamic system**: its state changes over time as leaders launch experiments, deploy models, modify processes, reallocate resources and respond to market signals.

It is also a **learning system**, because those actions generate outcomes and those outcomes should feed back into future decisions. In this sense, MLOps is not only a technical discipline for deploying models; it is part of the enterprise feedback mechanism that allows the organization to observe, act, measure and adapt.

3.1 A formal model of the enterprise

An enterprise $\mathcal{E}$ can be viewed as a dynamic system composed of capabilities $\mathcal{C}$, processes $\mathcal{P}$, resources $\mathcal{R}$, data assets $\mathcal{D}$ and decision mechanisms $\mathcal{M}$. In the context of MLOps and Data Science, the decision mechanisms of interest are machine learning models; for the purposes of this discussion we therefore denote the set of all ML models by $\mathcal{M}$.

Let $\mathcal{S}$ be the set of possible enterprise states, let $\mathcal{A}$ be the set of possible organizational actions, and let $t \in \mathbb{N} \cup \{0\}$ denote discrete time. A **state-transition function** is a map

$$F: \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S},$$

such that

$$S_{t+1} = F(S_t, A_t).$$